AMS Design Course

Information Technology Institute — ITI

Lab 8 Report

PAD Ring Design | LVS & ERC | LEF Generation

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Task 1 — ERC & PAD Rule Fixes

1.1 Rule File Modifications

The following changes were made to the DRC/ERC rule file:

- Added pad layer stack rules to allow AP layer contacts on the bond pad area.
- Adjusted antenna ratio limits for PAD cells to match foundry requirements.
- Modified spacing rules between PAD and seal ring layers.
- Excluded internal ESD layers from standard metal spacing checks inside pad cells.

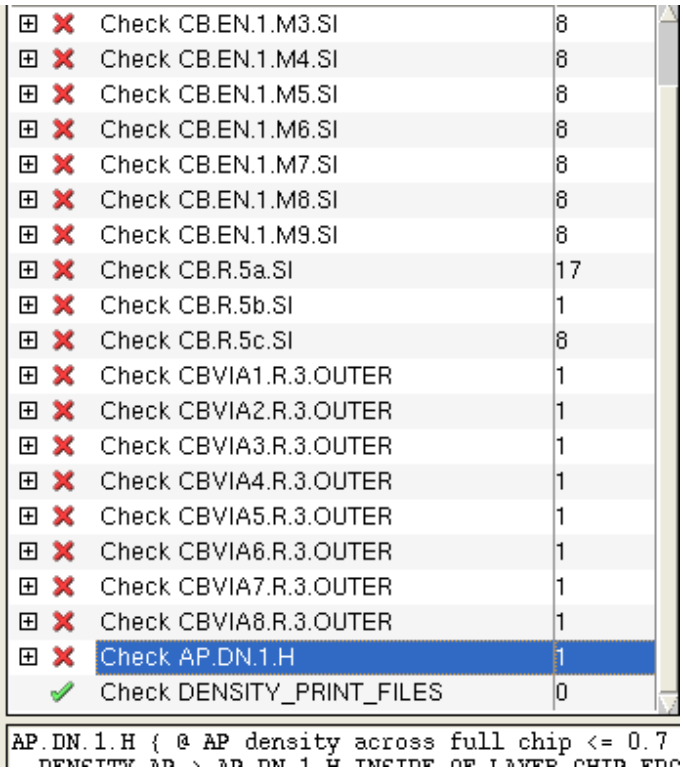
1.2 Rule File — Modifications Screenshot

```
single_in_line pad ONLY
//#DEFINE PITCH_60_SINGLE        //Turn on to use IO cell (>=60um pitch) of the
single_in_line pad ONLY

//#DEFINE PITCH_45_STAGGER        //Turn on to use IO cell (>=45um pitch) of the
 staggered pad ONLY
//#DEFINE PITCH_50_STAGGER        //Turn on to use IO cell (>=50um pitch) of the
 staggered pad ONLY
//#DEFINE PITCH_60_STAGGER       //Turn on to use IO cell (>=60um pitch) of the
staggered pad ONLY
//#DEFINE PITCH_70_STAGGER       //Turn on to use IO cell (>=70um pitch) of the
staggered pad ONLY
//#DEFINE PITCH_80_STAGGER       //Turn on to use IO cell (>=80um pitch) of the
staggered pad ONLY

//#DEFINE PITCH_80_TRITIER       //Turn on to use IO cell (>=80um pitch) of the
tri_tier pad ONLY
/* Please note that only one kind of pitch can exist for each pad structure (sir
gle_in_line, stagger, tri_tier) */
```

Fig 1 — Rule file modifications (commented/changed rules)

1.3 WB Violations — lab08/PAD

Screenshots below show the triggered WB violations and the PAD layout:



⊞ ✗	Check CB.EN.1.M3.SI	8
⊞ ✗	Check CB.EN.1.M4.SI	8
⊞ ✗	Check CB.EN.1.M5.SI	8
⊞ ✗	Check CB.EN.1.M6.SI	8
⊞ ✗	Check CB.EN.1.M7.SI	8
⊞ ✗	Check CB.EN.1.M8.SI	8
⊞ ✗	Check CB.EN.1.M9.SI	8
⊞ ✗	Check CB.R.5a.SI	17
⊞ ✗	Check CB.R.5b.SI	1
⊞ ✗	Check CB.R.5c.SI	8
⊞ ✗	Check CBVIA1.R.3.OUTER	1
⊞ ✗	Check CBVIA2.R.3.OUTER	1
⊞ ✗	Check CBVIA3.R.3.OUTER	1
⊞ ✗	Check CBVIA4.R.3.OUTER	1
⊞ ✗	Check CBVIA5.R.3.OUTER	1
⊞ ✗	Check CBVIA6.R.3.OUTER	1
⊞ ✗	Check CBVIA7.R.3.OUTER	1
⊞ ✗	Check CBVIA8.R.3.OUTER	1
⊞ ✗	Check AP.DN.1.H	1
✓	Check DENSITY_PRINT_FILES	0

```
AP.DN.1.H { @ AP density across full chip <= 0.7
```

Fig 2 — WB violations triggered in lab08/PAD

⊞ ✗	Check CB.EN.1.M3.SI	8
⊞ ✗	Check CB.EN.1.M4.SI	8
⊞ ✗	Check CB.EN.1.M5.SI	8
⊞ ✗	Check CB.EN.1.M6.SI	8
⊞ ✗	Check CB.EN.1.M7.SI	8
⊞ ✗	Check CB.EN.1.M8.SI	8
⊞ ✗	Check CB.EN.1.M9.SI	8
⊞ ✗	Check CB.R.5a.SI	17
⊞ ✗	Check CB.R.5b.SI	1
⊞ ✗	Check CB.R.5c.SI	8
⊞ ✗	Check CBVIA1.R.3.OUTER	1
⊞ ✗	Check CBVIA2.R.3.OUTER	1
⊞ ✗	Check CBVIA3.R.3.OUTER	1
⊞ ✗	Check CBVIA4.R.3.OUTER	1
⊞ ✗	Check CBVIA5.R.3.OUTER	1
⊞ ✗	Check CBVIA6.R.3.OUTER	1
⊞ ✗	Check CBVIA7.R.3.OUTER	1
⊞ ✗	Check CBVIA8.R.3.OUTER	1
✓	Check DENSITY_PRINT_FILES	0

```
CB.EN.1.M4.SI { @ CB enclosure by M4 pad regions >= 1.5um
  ENC CB_SINGLE_NON_CUP M4 < 1.5 ABUT <90 SINGULAR REGION
  CB_SINGLE_NON_CUP OUTSIDE EDGE M4
```

Fig 3 — lab08/PAD layout after solving violations

1.4 WB Violations — lab08/padring

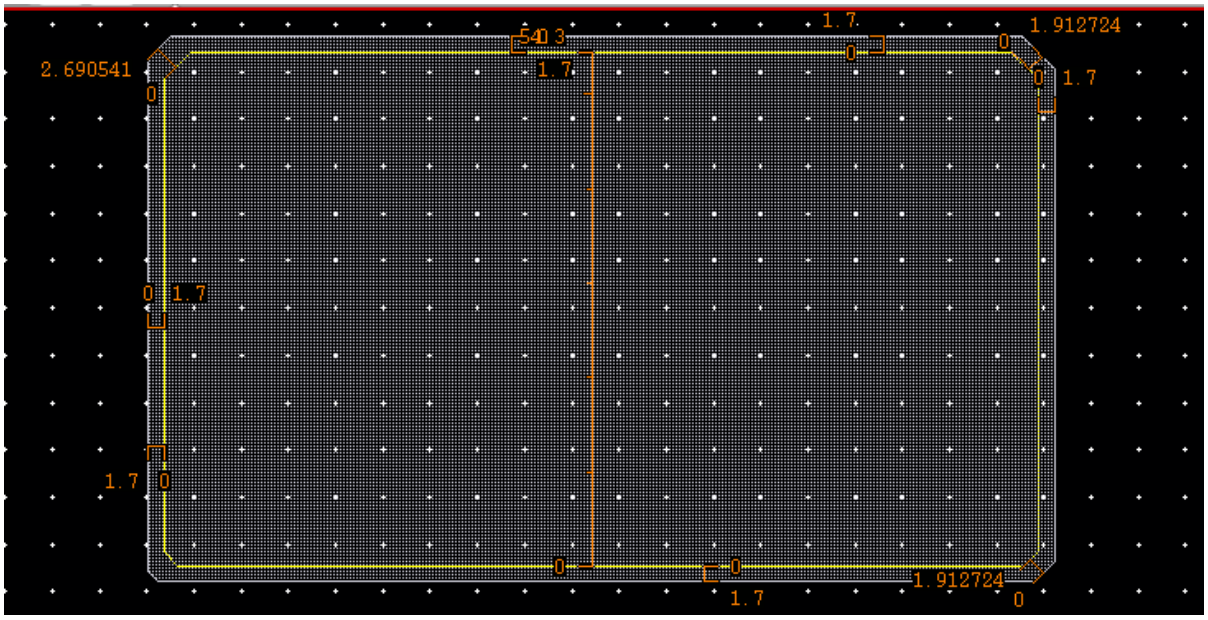


Fig 4 — lab08/padring layout

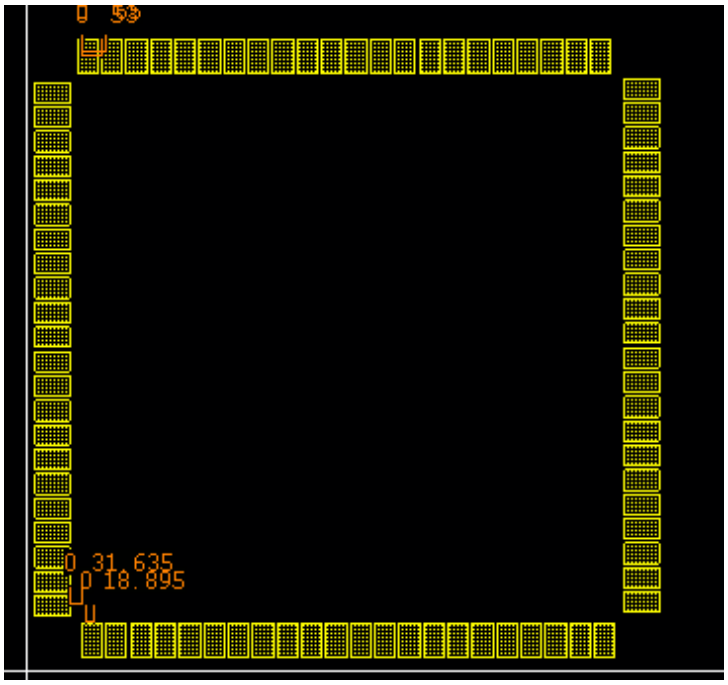


Fig 5 — lab08/padring layout after solving violations

Task 2 — Floorplan, ESD Resistance & LVS

2.1 Floorplan

Block areas used for floorplanning:

Cell	Area (µm²)	Width (µm)	Height (µm)
LDO	307,076	997	308
BandGap (BG)	181,597	793	229
DFF	70	10	7
Internal Circuit	250,000	500	500
Internal Circuit2	250,000	500	500
DIG	480,000	350	1374

Available routing area: 1700 × 1700 − (480,000 + 250,000×2 + 181,597 + 307,076 + 70) = 1,059,357 µm²

Floorplan Layout

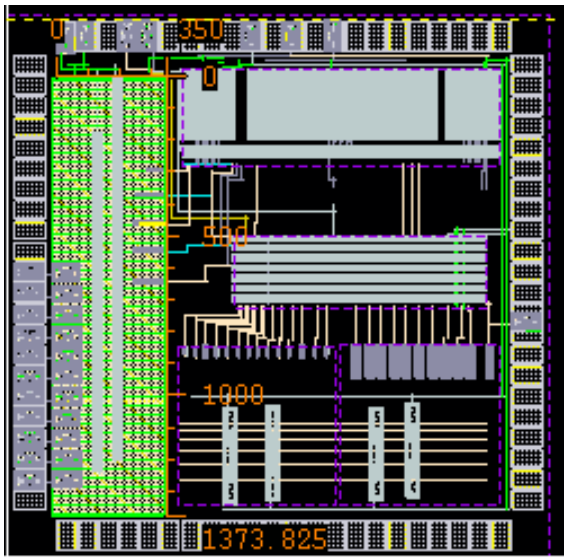


Fig 6 — Floorplan overview

Pin Placement Strategy

Input pins placed on the left side — digital block on the left, output pins on the right to route into the circuits.

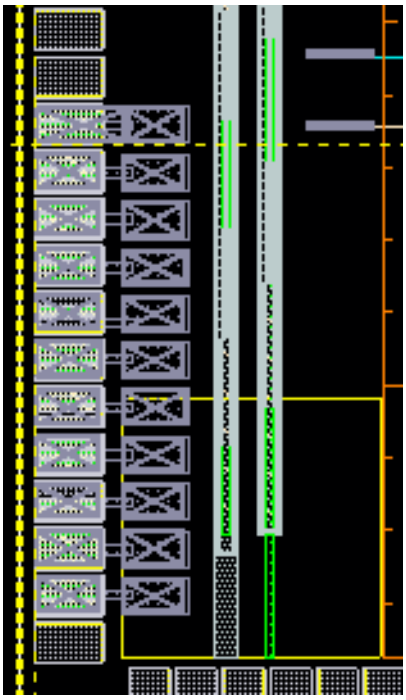


Fig 7 — Input pin placement (left side)

FF pins placed at the top, as close as possible to FF and PCLAMP, to reduce wire resistance on the ESD path.

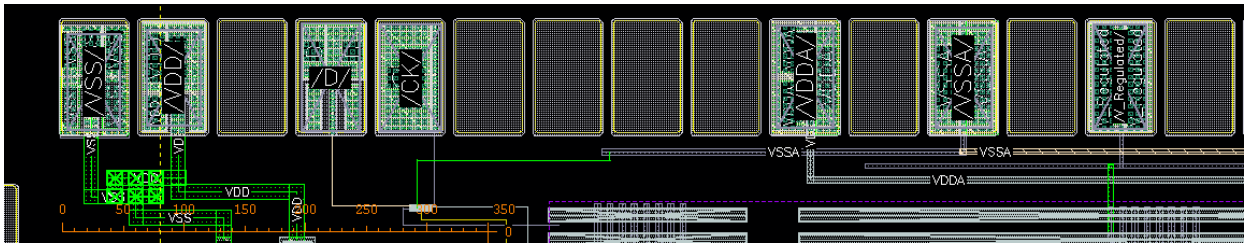


Fig 8 — FF pin placement at top

Output pin placed on the right near the Internal Circuit output to reduce resistance.

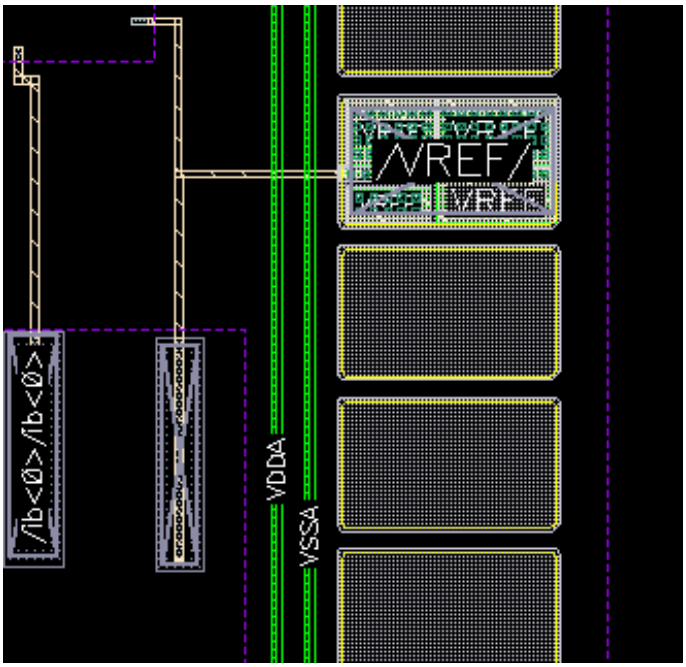


Fig 9 — Output pin placement (right side)

2.2 ESD Resistance Measurements

Point-to-point (P2P) resistance was measured for two ESD discharge paths:

- GND pad (AP layer) → Pclamp_replacement (M7)
- SPLY pad (AP layer) → Pclamp_replacement (M7)

Resistance on VDDA:

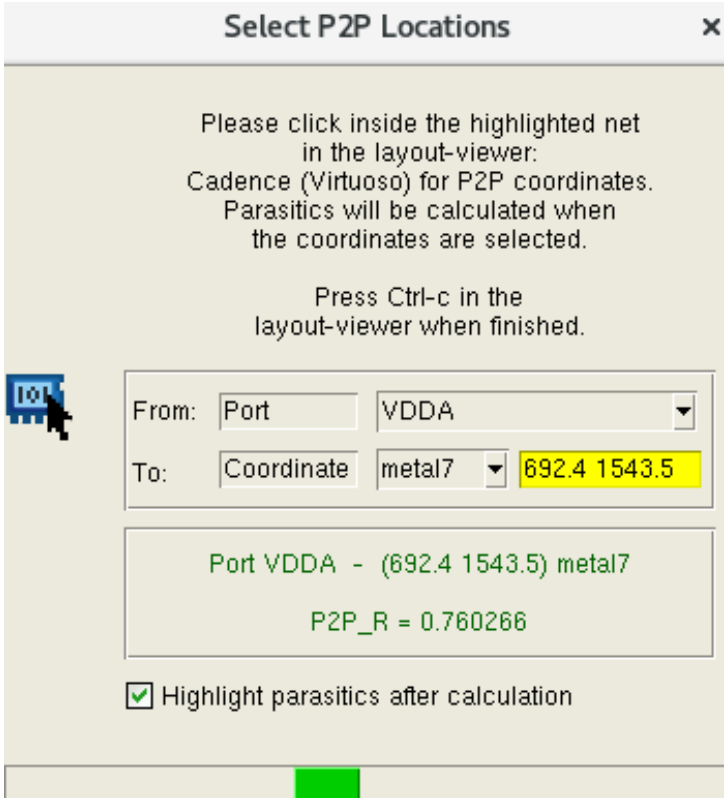


Fig 10 — P2P resistance measurement (VDDA path)

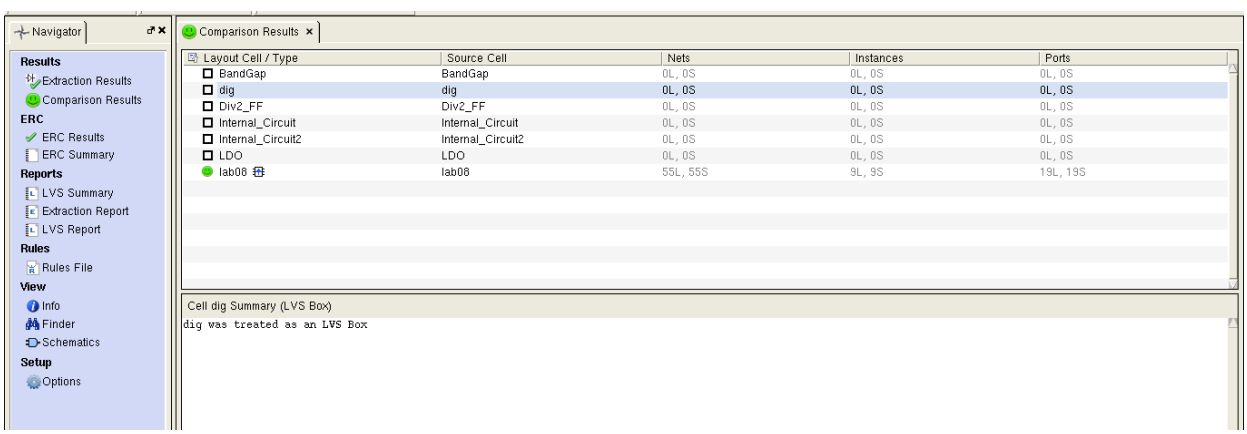


Fig 11 — P2P resistance result

2.3 LVS Settings

LVS Box options configured as shown below:

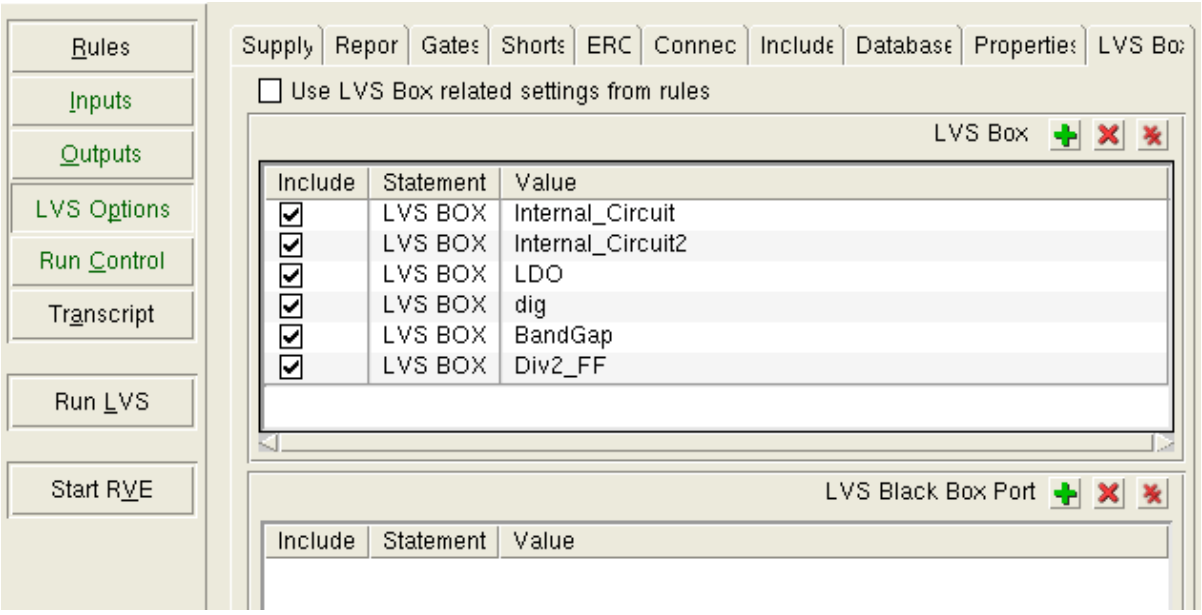


Fig 12 — LVS Box options (1)

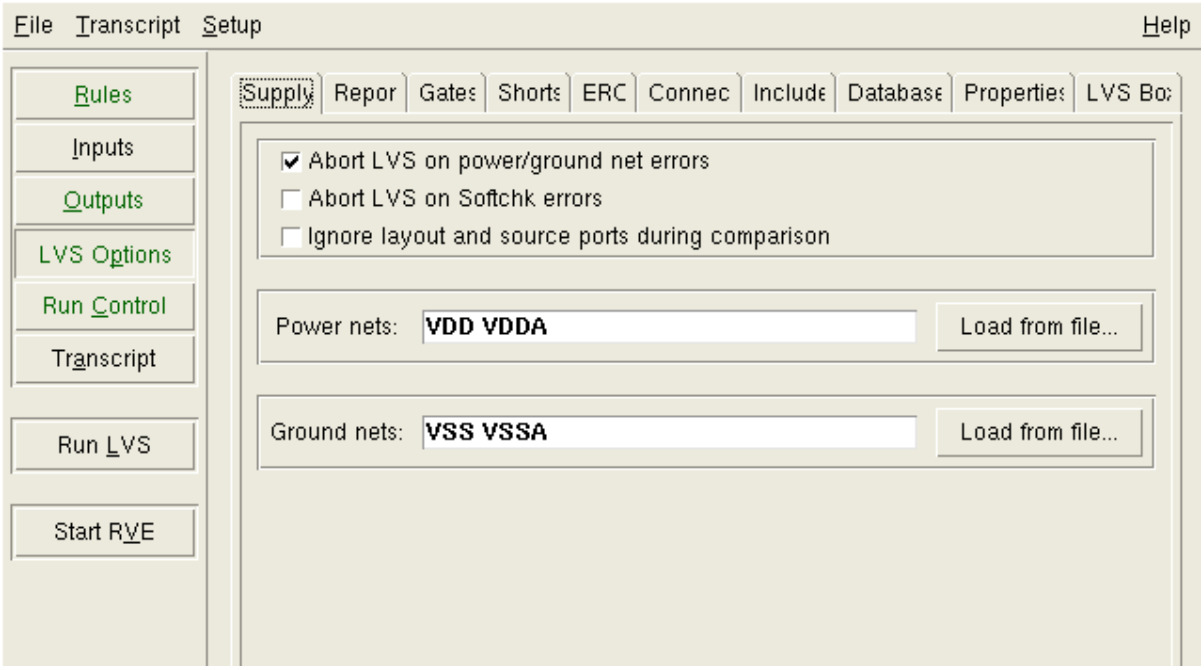


Fig 13 — LVS Box options (2)

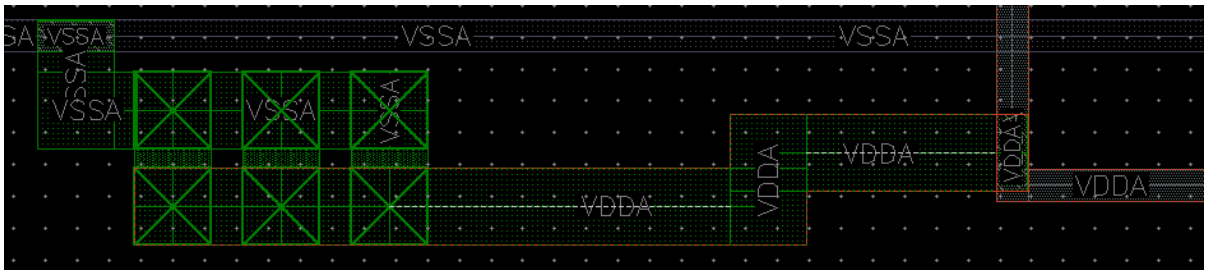


Fig 14 — LVS Box options (3)

2.4 Additional Layers Added to Pass LVS

[Add any extra layers you added to the layout here, with a short reason for each]

2.5 Challenges Faced

1. Connecting pins on pads was hard and time-consuming — required careful LVS Box setup to let the tool discover the correct net names.
2. Connection to PCLAMP needed routing on a higher metal layer with minimum length to keep resistance within ESD budget.
3. Floorplanning BG, LDO, and Internal Circuits within the fixed die size required several tries to fit all blocks correctly.

2.6 Drive Links

File / Folder	Drive Link
lab08 library	[paste link here]
LVS report (.lvs.report)	[paste link here]
ERC summary (.erc.summary)	[paste link here]

Task 3 — LEF Generation

3.1 Digital Layout — Pin Distribution

Input pins are on the left, output pins on the right, VDD and VSS in the middle.

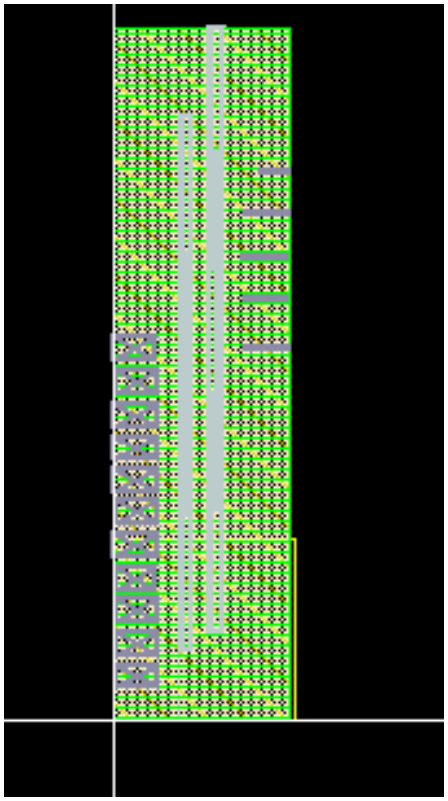


Fig 15 — Digital layout pin distribution

3.2 LEF File Content

```
VERSION 5.8 ;

BUSBITCHARS "[]" ;

DIVIDERCHAR "/" ;

PROPERTYDEFINITIONS

MACRO lastSavedExtractCounter INTEGER ;

END PROPERTYDEFINITIONS

MACRO dig

ORIGIN 0 0 ;

FOREIGN dig 0 0 ;
```

```
SIZE 357.77 BY 357.77 ;

PIN PAD_input[10]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 4.55 66.115 87.15 108.915 ;

END

END PAD_input[10]

PIN PAD_input[1]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 5.55 648.215 87.05 692.215 ;

END

END PAD_input[1]

PIN PAD_input[0]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 2.95 712.315 86.15 757.815 ;

END

END PAD_input[0]

PIN PAD_input[2]

DIRECTION INPUT ;

USE SIGNAL ;
```

```
PORT
LAYER M4 ;
RECT 3.95 581.315 87.65 626.315 ;
END
END PAD_input[2]
PIN PAD_input[3]
DIRECTION INPUT ;
USE SIGNAL ;
PORT
LAYER M4 ;
RECT 3.65 516.315 88.65 560.615 ;
END
END PAD_input[3]
PIN PAD_input[4]
DIRECTION INPUT ;
USE SIGNAL ;
PORT
LAYER M4 ;
RECT 3.95 452.615 87.45 497.615 ;
END
END PAD_input[4]
PIN PAD_input[5]
DIRECTION INPUT ;
USE SIGNAL ;
PORT
LAYER M4 ;
RECT 4.45 389.615 87.65 434.315 ;
END
```

```
END PAD_input[5]

PIN PAD_input[6]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 3.95 323.915 88.25 368.415 ;

END

END PAD_input[6]

PIN PAD_input[7]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 4.95 258.615 85.95 302.415 ;

END

END PAD_input[7]

PIN PAD_input[8]

DIRECTION INPUT ;

USE SIGNAL ;

PORT

LAYER M4 ;

RECT 4.45 194.415 86.45 240.115 ;

END

END PAD_input[8]

PIN PAD_input[9]

DIRECTION INPUT ;

USE SIGNAL ;
```

```
PORT
LAYER M4 ;
RECT 4.75 128.915 86.95 173.415 ;
END
END PAD_input[9]
PIN VSS
DIRECTION INPUT ;
USE SIGNAL ;
NETEXPR "VSS VSS" ;
PORT
LAYER M5 ;
RECT 133.55 141.915 157.55 1195.015 ;
END
END VSS
PIN VDD
DIRECTION INPUT ;
USE SIGNAL ;
NETEXPR "VDD VDD" ;
PORT
LAYER M5 ;
RECT 193.55 174.015 217.05 1367.615 ;
END
END VDD
PIN ldo_bypass
DIRECTION OUTPUT ;
USE SIGNAL ;
NETEXPR "ldo_bypass ldo_bypass" ;
PORT
```

```
LAYER M4 ;
RECT 292.65 1080.815 349.65 1084.015 ;
END
END ldo_bypass
PIN ldo_ok
DIRECTION OUTPUT ;
USE SIGNAL ;
NETEXPR "ldo_ok ldo_ok" ;
PORT
LAYER M4 ;
RECT 260.25 1001.315 348.75 1004.615 ;
END
END ldo_ok
PIN bg_ok
DIRECTION OUTPUT ;
USE SIGNAL ;
NETEXPR "bg_ok bg_ok" ;
PORT
LAYER M4 ;
RECT 261.05 828.715 346.25 834.615 ;
END
END bg_ok
PIN por
DIRECTION OUTPUT ;
USE SIGNAL ;
NETEXPR "por por" ;
PORT
LAYER M4 ;
```

```
RECT 260.05 913.615 346.95 917.215 ;

END

END por

PIN pwr_ok

DIRECTION OUTPUT ;

USE SIGNAL ;

NETEXPR "pwr_ok pwr_ok" ;

PORT

LAYER M4 ;

RECT 260.25 731.115 347.75 737.615 ;

END

END pwr_ok

OBS

LAYER M9 ;

RECT 1.59 0 350.07 1367.615 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER VIA8 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER M8 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER VIA7 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER M7 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER VIA6 ;

RECT 1.59 0 350.07 1367.615 ;

LAYER M6 ;

RECT 1.59 0 350.07 1367.615 ;
```

```
END

PROPERTY lastSavedExtractCounter 471 ;

END dig

END LIBRARY
```

3.3 LEF File Drive Link

https://drive.google.com/drive/folders/1nHVgqkn7Wx66n6WJjyW42z8jp2Csv4LB?usp=sharing